Tissue-integrated bionic knee restores versatile legged movement after amputation

Tony Shu; Daniel Levine; Seong Ho Yeon; Ethan Chun; Christopher C. Shallal; John McCullough; Rickard Brånemark; Matthew J. Carty; Marco Ferrone; Sean Boerhout; Alexander Ko; Corey L. Sullivan; Gloria Zhu; Michael Nawrot; Matthew Carney; Ged Wieschhoff; Gabriel Friedman; Hugh Herr

Science
DOI: 10.1126/science.adv3223

Abstract

Lower-extremity prostheses have evolved through mechanical redesigns that prioritize improved cyclic locomotion. However, this limited approach to limb restoration has precluded necessary progress toward recovering the versatile acyclic movements that constitute the remainder of human athleticism. We present an osseointegrated mechanoneural prosthesis that incorporates modified hard and soft tissues along with permanently implanted hardware in a neuroembodied design. We developed a biomimetic coupling between neuromuscular signaling and joint movement that exceeds the versatility of established control methods, which depend upon conventional amputation musculature and surface electromyography. Our findings also reveal that superior residual neuromuscular function can enable prosthetic movement speeds surpassing that of intact physiology. Anatomical prosthetic integration may be necessary for meeting, and possibly exceeding, the movement capabilities of an intact limb.